

P4-16 are  
5 V. Select a  
variation mode

re P4-18 are  
r  $v_S = 0.8$  V.

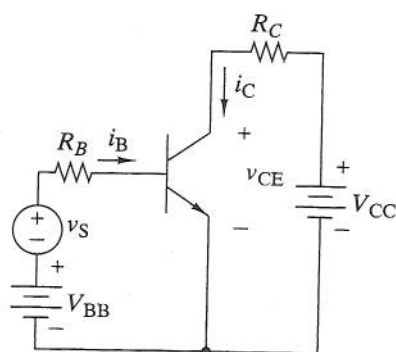


FIGURE P4-20

### OBJECTIVE 4-3 OP AMP CIRCUIT ANALYSIS (SECTS. 4-4, 4-5)

Given a circuit consisting of linear resistors, OP AMPs, and independent sources, find selected output signals or input-output relationships.

See Examples 4-13, 4-14, 4-16, 4-17, 4-18, 4-19 and Exercises 4-10, 4-11, 4-12, 4-13, 4-15

are  $\beta = 150$

4-21 Find  $v_O$  in terms of  $v_S$  in Figure P4-21.

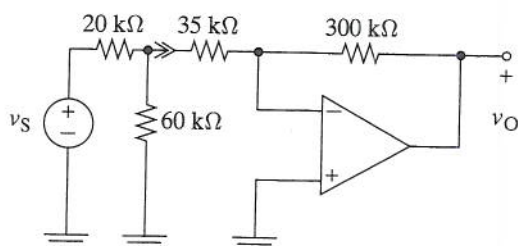


FIGURE P4-21

4-22 What is the range of the gain  $v_O/v_S$  in Figure P4-22?

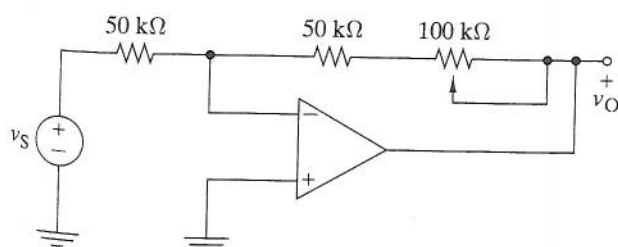


FIGURE P4-22

4-23 (a) Find  $v_O$  in terms of  $v_S$  in Figure P4-23.  
(b) Find  $i_O$  for  $v_S = 1.5$  V.

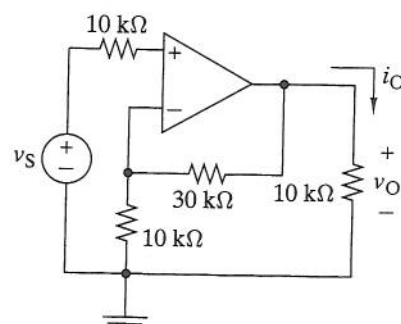


FIGURE P4-23

4-24 (a) Find  $v_O$  in terms of  $v_S$  in Figure P4-24.  
(b) Find  $i_O$  for  $v_S = 2$  V.

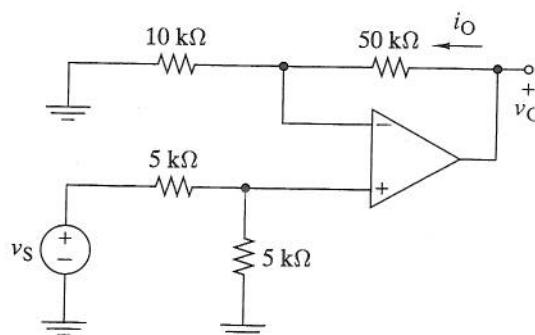


FIGURE P4-24

4-25 What is the gain  $v_O/v_S$  in Figure P4-25 when the switch is in position 1? Repeat for positions 2 and 3.

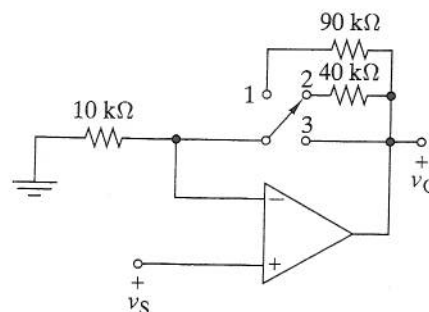


FIGURE P4-25

4-26 Find  $v_O$  in terms of the inputs  $v_1$  and  $v_2$  in Figure P4-26.

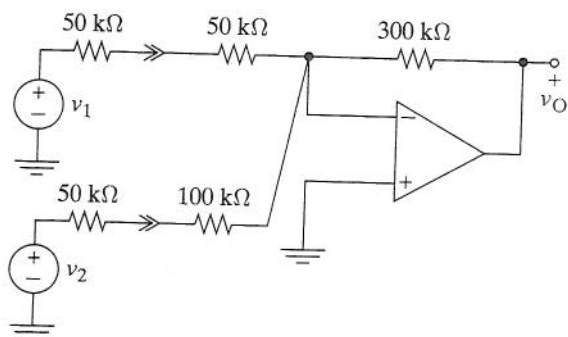


FIGURE P4-28

- 4-27 The input-output relationship for a three-input inverting summer is

$$v_O = -(v_1 + 2v_2 + 5v_3)$$

The resistance of the feedback resistor is  $50 \text{ k}\Omega$  and the supply voltages are  $V_{CC} = \pm 15 \text{ V}$ .

- (a) Find the values of the input resistors  $R_1$ ,  $R_2$ , and  $R_3$ .  
 (b) For  $v_2 = 0.5 \text{ V}$  and  $v_3 = -1 \text{ V}$ , find the allowable range of  $v_1$  for linear operation.

- 4-28 Find  $v_O$  in terms of the inputs  $v_1$  and  $v_2$  in Figure P4-28.

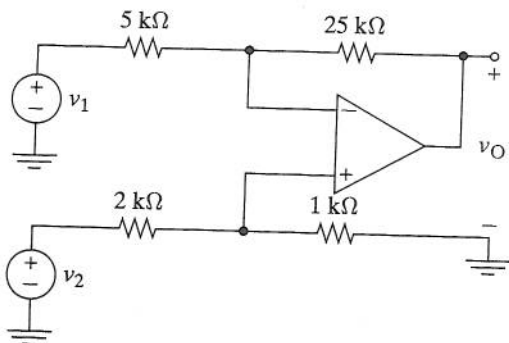


FIGURE P4-28

- 4-29 Find  $v_O$  in terms of the inputs  $v_{S1}$  and  $v_{S2}$  in Figure P4-29.

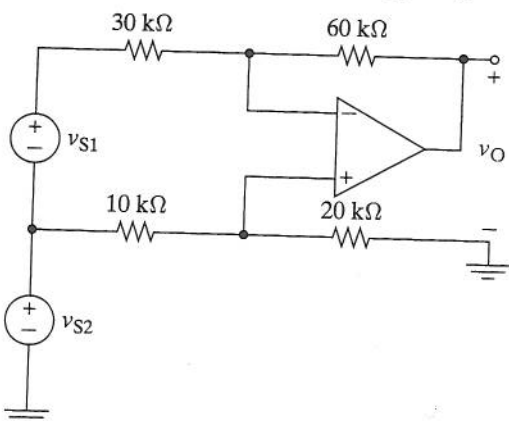


FIGURE P4-29

- 4-30 Find  $v_O$  in terms of the inputs  $v_{S1}$  and  $v_{S2}$  in Figure P4-30.

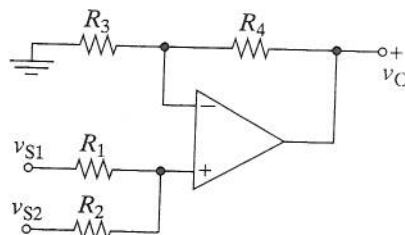


FIGURE P4-30

- 4-31 It is claimed that  $v_O = v_S$  when the switch is closed in Figure P4-31 and that  $v_O = -v_S$  when the switch is open. Prove or disprove this claim.

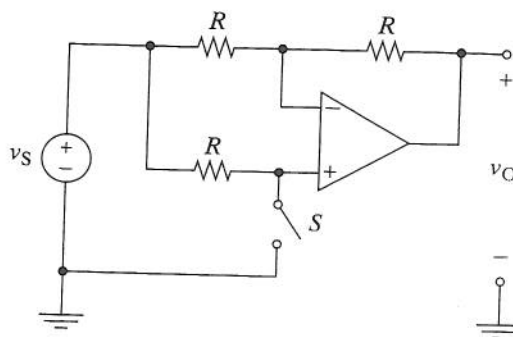


FIGURE P4-31

- 4-32 The inputs to the circuit in Figure P4-32 are a current source  $i_S$  and a voltage source  $v_S$ . When the OP AMP is in its linear range, the output voltage has the form  $v_O = K_1 v_S + K_2 i_S$ . Find the constants  $K_1$  and  $K_2$ .

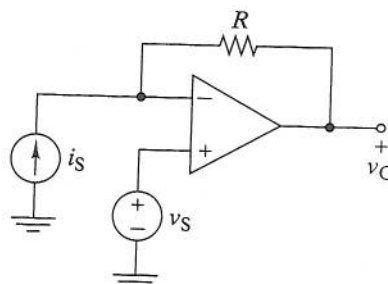


FIGURE P4-32

- 4-33 Use node-voltage analysis to show that the input-output relationship of the circuit in Figure P4-33 is

$$v_O = -\left(\frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R R_2}\right) v_S$$

4-34 Use n  
 $v_O = -v_S$   
 circuit is

o-  
 $v_S$

F I

4-35 Find  $v_C$   
 P4-35.

$v_{S1} +$

$v_{S2} +$

$v_{S3} +$

FIGU

4-36 Find  $v_O$  i

$v_{S1} +$   $R_1$

FIGURE I



in Figure P4-30.

$v_O$

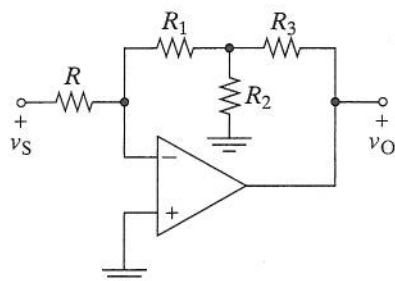


FIGURE P4-33

4-34 Use node-voltage analysis in Figure P4-34 to show that  $i_O = -v_S/2R$  regardless of the load. That is, show that the circuit is a voltage-controlled current source.

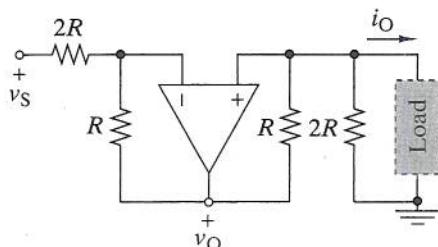


FIGURE P4-34

4-35 Find  $v_O$  in terms of the inputs  $v_{S1}$ ,  $v_{S2}$ , and  $v_{S3}$  in Figure P4-35.

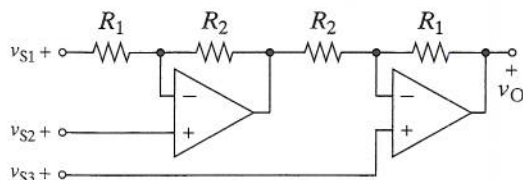


FIGURE P4-35

4-36 Find  $v_O$  in terms of the inputs  $v_{S1}$  and  $v_{S2}$  in Figure P4-36.

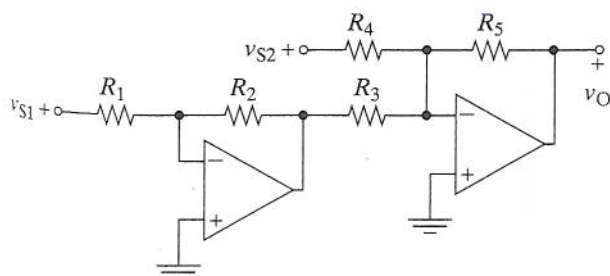


FIGURE P4-36

4-37 Find the output  $v_2$  in terms of the input  $v_1$  in Figure P4-37.

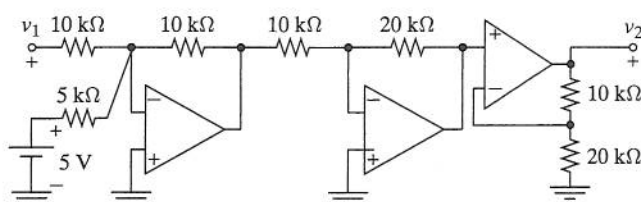


FIGURE P4-37

4-38 Find the output  $v_2$  in terms of the input  $v_1$  in Figure P4-38.

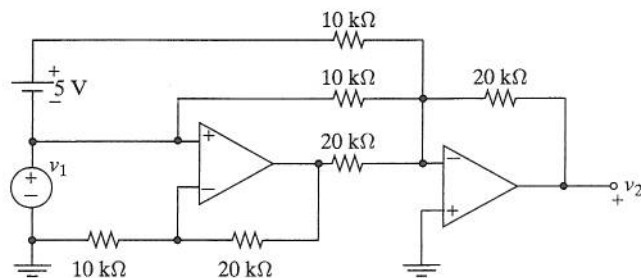


FIGURE P4-38

4-39 Find the output  $v_O$  in terms of the input  $v_S$  in Figure P4-39.

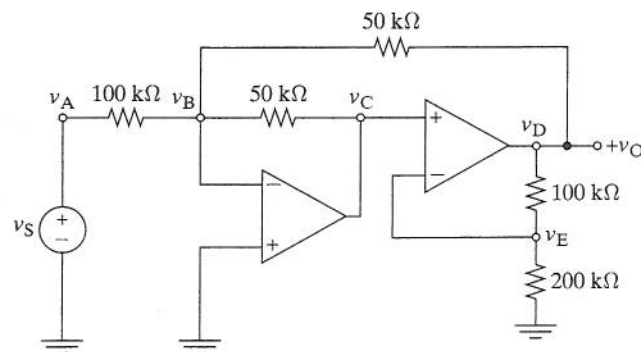


FIGURE P4-39

4-40 Find the output  $v_O$  in terms of the inputs  $v_1$  and  $v_2$  in Figure P4-40.